

Quiz 2

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (4 points) Given two matrices $A = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -4 & 0 \\ -1 & 3 & 1 \end{bmatrix}$.

Compute AB .

$$AB = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 3 & -4 & 0 \\ -1 & 3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & 3 \\ 0 & -10 & -9 \\ 0 & 5 & 4 \end{bmatrix}$$

Problem 2. (6 points) Write down the 3×3 matrices that produce the following operations:

(a) E_{21} subtracts $1/3$ times row 1 from row 2.

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{3} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) E_{32} adds 5 times row 2 to row 3.

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix}$$

(c) E_{31} such that $E_{31} \begin{bmatrix} 2 \\ 3 \\ -4 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$